

MANISH PATIYAL

Principal Technical Lead | Solution Architect | AI Augmented Engineer

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| Mandi, Himachal Pradesh

PROFESSIONAL SUMMARY

Principal Technical Lead with 12+ years of experience building scalable mobile applications across Android (Kotlin, Jetpack, MVVM), React Native, iOS, and Flutter. Proven leader in architecture, performance optimization, and engineering excellence, with expertise in React.js, Node.js, GraphQL, PostgreSQL, and MongoDB. Delivered products with 5M+ downloads, 98%+ crash-free stability, 40%+ performance improvements, and 50% faster deployments.

CORE COMPETENCIES

Programming Languages: Kotlin, Java, JavaScript, TypeScript, Python

Android: Jetpack Compose, Coroutines, Flow, LiveData, ViewModel, Room, Hilt, WorkManager, MVVM, Clean Architecture, Memory Profiling, ANR Debugging, Perfetto

Cross-Platform: React Native, Flutter, iOS

APIs & Data: GraphQL, Apollo Client, REST APIs, PostgreSQL, MongoDB, Offline-First Sync, Real-Time Data

DevOps & Tools: Docker, Bitrise, Jenkins, GitHub Actions, Firebase

Leadership: Architecture Design, Team Mentoring, Code Reviews, Code Quality Standards, Performance Optimization, Release Management

PROFESSIONAL EXPERIENCE

Senior Principal Technical Lead / Senior Software Engineer

VT Netzwelt Pvt. Ltd., Chandigarh | May 2017 - Present

- Architected and scaled mobile solutions for global enterprise clients using Android, React Native, and iOS; led team of 5+ engineers delivering production-grade applications
- Achieved 40%+ performance improvements across platforms: optimized startup time, reduced memory footprint, eliminated ANRs; designed component library (50+ components) supporting diverse device types

- Mentored 8+ developers on architecture and code quality; optimized CI/CD pipelines (50% faster deployments, bi-weekly releases); managed complex third-party SDK integrations

Software Engineer

IDEMIA, Noida | Aug 2013 – May 2017

- Developed Android biometric solutions integrating proprietary SDKs for healthcare, telecom, and security sectors; architected Remote SIM Personalization platform for enterprise tablets with complex hardware-software integration and secure provisioning; designed reusable components for biometric enrollment, authentication, and identity verification with hands-on enterprise customer support
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EDUCATION

Bachelor of Technology (B-Tech) in Computer Science & Engineering

Lovely Professional University, Jalandhar | 2013

- CGPA: 8.75/10
- Relevant coursework: Data Structures, Algorithms, Operating Systems, Database Management, Software Engineering, Artificial Intelligence

Diploma in Computer Science

Government Polytechnic, Sundernagar | 2010

KEY ACHIEVEMENTS & PROJECTS

GfK Consumer Scan — Large-Scale Mobile Application (5M+ Downloads)

- Architected production-grade React Native application from conception to massive scale
- Implemented clean MVVM architecture with LiveData for state management
- Engineered data layer using Room ORM with optimized queries for offline first synchronization
- Designed delta sync algorithm to minimize network bandwidth
- Integrated barcode scanning (Camera2 API), image processing pipelines, REST/GraphQL APIs
- Used WorkManager for reliable background synchronization
- Achieved 98% crash-free rate through Jest testing (300+ unit tests), integration tests, and continuous monitoring
- Mentored team of 8+ developers on architectural decisions and best practices
- Maintained consistent sub 100ms UI response times through performance optimization

Keyhook — 360° Real Estate Visualization Platform

- Designed React Native + Native bridge architecture for complex Insta360 Camera SDK integration; demonstrated expertise in system design at scale, concurrency, threading models, and memory optimization
 - Built custom native modules exposing camera APIs to JavaScript layer with proper lifecycle management; implemented real time image processing and media streaming at 60fps with zero data loss
 - Delivered end-to-end solution from architecture through production deployment
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